batmonitor 0.74: discharge path of one cycle (bm_step, every 5 s) | max persistent decision states 12288 = D1 2 x D2 128 x lead 3 x D3 2 x D4 2 x D7 4 | blue = decision (c/bm_tables.h, make check) | green = discharge matrix | [CBMC] = proved for all inputs

**inputs** (chain_inputs)
PCC (live), Sofar Bat1 W + SoC, heat pump W (SDM72D), PV,
BMS SoC / U / CCL / DCL / cells, AC-in / AC-out per phase
-> stale, wp_on (fresh and ≥ WP_ON_TH 300 W), night (PV < NIGHT_PV_TH 100 W), winter (Oct..Mar)

**D7** house on AC-out1 (0.69-c) state: house_on, under timer
HOUSE_ACOUT 0 off / 1 on / 2 auto (on > 99 W, off after 60 s < 30 W)
on: no source row, no night floor, no heat pump cap (src text Z1 / Z2)
[make check: 54 combinations, 29 on, 4 states reachable] Lenggries env: HOUSE_ACOUT=0

**D1** bm_d1 state: bat1_first
Sofar first: Bat1 SoC > BAT1_SOC_MIN (on again from +2 %) and heat pump on
winter heat pump rule: winter and heat pump on -> Bat1 discharging: 0 W, else WP_WINTER_W
[make check: 128 cells, 14 outcomes; details wp_matrix]

src

WPMAX

**S3 source**
day -> **DAY**
night + Sofar first -> **B1**
winter night + heat pump on -> **WPG** (no night floor)
else night -> **STK** (night floor)

**house** = PCC + own charge (the real PCC, target 0 W)
dis_max = W_MAX, or the winter heat pump cap (rule text WPMAX(W))

**S0 gates** per bank (chain_gates)
SoC from the BMS; caps per phase, one per bank:
chg_cap = min(charger x U, BMS CCL, top hold) dis_cap = min(DISCHARGE_MAX_PHASE, BMS DCL)

surplus

**D2** bm_d2 per bank state: prot, force, low, chg_prev
first match: BMS_MISSING > FORCE (force or low) > STALE > LADESPERRE >
CHARGE / FULL > HOLD > PROT > CHARGING > **DIS**
**LOW**: on at DCL < DCL_FORCE_A 1 A or lowest cell < CELL_FORCE_V,
off when DCL is back, the cell ≥ CELL_FORCE_RELEASE_V and SoC ≥ SOC_MIN_RELEASE;
LOW -> FORCE (charge, no discharge); Lenggries env: CELL_FORCE_V 3.05 V / release 3.20 V
HOLD: the force_full_from_pv stack never discharges
make check: 64512 combinations per stack -> DIS 360 of them
[CBMC h_d2_bits; threshold order needed: h_d2_bits_any_order]

**D-lead** bm_dlead state: lead
|SoC1 - SoC2| > SOC_BALANCE_ON -> the stack ahead leads
< SOC_BALANCE_OFF -> no lead

row

house

lead

dis_cap

mode per phase

DIS

else

CHARGE

**discharge matrix** (0.45-c, whole watts)

**BM_SRC_RULE[src]**

| SRC | b1_signed | trickle | idle_w | prop_hold |
|---|---|---|---|---|
| DAY | 0 | 0 | 20 | 0 |
| B1 | 0 | 0 | 20 | 0 |
| STK | 1 | 1 | 0 | 1 |
| WPG | 0 | 0 | 20 | 0 |

**deficit** (int W)
b1 = b1_signed ? Bat1 : max(Bat1 charge, 0)
target = SOYO_TARGET + (trickle ? SOFAR_TRICKLE : 0)
deficit = own_discharge - b1 - KP_PM x (house - target) / 1000

D7 on

**D3** bm_d3 state: prop_prev
PROP = import (< -100 W) or
((prop_prev or prop_hold) and deficit > 20 W)

**D7 on** (no PROP / IDLE state)
winter: w = AC-out1 load (Z2 near 0) -> **HOUSE**
summer: w = max(PCC deficit, AC-out1 load) -> **PROP** / **HOUSE**
winter export cap (0.74-c): w ≤ what keeps the export at CHARGE_RESERVE -> **EXPCAP**
at most W_MAX

**dis_amount**: w = min(PROP ? max(deficit, 0) : idle_w, dis_max)
[CBMC h_dis_amount]

**alloc_discharge** on **alloc_levels** (one split core for charge and discharge)
always all discharging phases (single-unit D6 / 1PH removed in 0.70-c)
lead: level 0 = lead stack (BAL), level 1 = only what its caps cut (SPILL / WAIT)
no lead: one level, every stack the same share (EQ); water filling up to dis_cap
[CBMC h_alloc_discharge: within caps, total ≤ w, lead first, equal per stack]

BMS_MISSING / FORCE / STALE / LADESPERRE /
FULL / HOLD / PROT / CHARGING
-> fixed setpoint (0, force charge incl. LOW, ...)

**charge** (chain_charge)
want = max(0, own_charge + KP x (surplus - CHARGE_RESERVE))
alloc_charge: CHARGE_PRIORITY, lead stack last, up to chg_cap
[CBMC h_charge]

**setpoints** per phase -> Multis (ESS AcPowerSetpoint)
rule text: <PROP / IDLE / HOUSE / EXPCAP> SRC:<DAY / B1 / STK / WPG / Z1 / Z2> <EQ / BAL / SPILL / WAIT> [WPMAX(W)] [DCL]
[CBMC h_discharge: whole discharge path]

PI prototype (shadow, not armed)